

App Configuration: Devicetree

Zephyr RTOS Course





Recap: L03 Kconfig



- `prj.conf` → `autoconf.h` → `CONFIG_*` macros in C
- `Kconfig` file in your app defines symbols
- `menuconfig` lets you browse and set options interactively
- Use `depends on`, hidden symbols, `choice`, `menu` to organize



Format



- Lecture with optional **live demo**
- Homework after each session (checked by us)
- Recordings will be shared
- Questions addressed at the end in a **Q&A session**



Where we left off



	Kconfig	Devicetree
Purpose	Software features	Hardware description
Question	"Do I want this feature?"	"Where is this hardware?"
Example	<code>CONFIG_LOG=y</code>	<code>zephyr,console = &uart0</code>
Files	<code>.conf</code> , <code>Kconfig</code>	<code>.dts</code> , <code>.overlay</code>
In code	<code>CONFIG_*</code> macros	<code>DT_*</code> macros



Devicetree



Devicetree



- Abstracts **board layout and peripherals**
- Describes what's on the board and how it's wired
- Independent of application logic



Devicetree: Syntax



Nodes, labels and properties



```
/ {  
    leds {  
        compatible = "gpio-leds"; // Tells Zephyr which driver to use  
  
        // node label: node name  
        // label - shortcut for referencing  
        // name - identifies node in tree  
        blue_led: led_name {  
            // &hsgpio0 - reference to another node  
            gpios = <&hsgpio0 0 GPIO_ACTIVE_LOW>;  
            label = "Blue LED"; // Key-value data  
        };  
    };  
};
```




Property value types



```
node {  
    // String  
    label = "Blue LED";  
    compatible = "gpio-leds";  
  
    // Integer (in angle brackets)  
    clock-frequency = <48000000>;  
  
    // Array of integers  
    reg = <0x40000000 0x1000>;  
  
    // Phandle + args (reference to another node)  
    gpios = <&hsgpio0 0 GPIO_ACTIVE_LOW>;  
  
    // Boolean (present = true, absent = false)  
    read-only;  
};
```



Devicetree: Overlays



What are overlays?



Patch files that modify the base board Devicetree – add new nodes or override existing properties.

```
/* boards/frdm_rw612.overlay */
/ {
    leds {
        blue_led: led_name {
            gpios = <&hsgpio0 0 GPIO_ACTIVE_LOW>;
        };
    };
};
```



Overlay loading order



Auto-discovery (when `DTC_OVERLAY_FILE` is not set):

1. `boards/<BOARD>.overlay` — matched by board name
2. `app.overlay` — fallback if no board overlay found

Stops at first match — if `boards/` overlay exists, `app.overlay` is ignored.



Override auto-discovery



Force a specific overlay:

```
west build -- -DDTC_OVERLAY_FILE="boards/frdm_rw612_red.overlay"
```

Add extra overlay on top of auto-discovered ones:

```
west build -- -DEXTRA_DTC_OVERLAY_FILE="my_extra.overlay"
```




Inspecting the Generated DTS



```
build/zephyr/zephyr.dts
```

- Final merged devicetree (board DTS + all overlays applied)
- Verify your overlay was correctly merged
- Find the actual node path for `DT_PATH()`
- Check that aliases point to the right nodes

FileEditSelectionViewGoRunTerminalHelp

ZEPHYR-COURSE

build

zephyr

kconfig

kernel

lib

misc

soc

subsys

.cmake.dotconfig.checksum

.config

.config.old

cmake_install.cmake

edt.pickle

edt.pickle.cmake

isr_tables_sw1.ld

isr_tables_vt.ld

isr_tables.c

libzephyr.a

linker_zephyr_pre0.cmd

linker_zephyr_pre0.cmd.dep

linker.cmd

linker.cmd.dep

runners.yaml

snippets_generated.cmake

syscall_weakdefs_llex.c

zephyr_final.map

zephyr_pre0.elf

zephyr_pre0.map

zephyr.bin

zephyr.dts

zephyr.dts.d

zephyr.dts.pre

zephyr.elf

zephyr.map

zephyr.stat

.ninja_deps

main.cpp 3

Kconfig U

zephyr.dts X

Kconfig.test U

FRDM-RW612-SCH.pdf U

build > zephyr > zephyr.dts

4 / {

10 chosen {

14 zephyr,hdrc-rcp-1t = &hdrc_rcp_1t; /* in deps/zephyr/dts/arm/nxp/nxp_rw6xx_common.dtsi:21 */

15 zephyr,sram = &sram_data; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:24 */

16 zephyr,flash = &w25q512jvfiq; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:25 */

17 zephyr,console = &flexcomm3; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:26 */

18 zephyr,shell-uart = &flexcomm3; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:27 */

19 zephyr,flash-controller = &w25q512jvfiq; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:28 */

20 zephyr,code-partition = &slot0_partition; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:29 */

21 zephyr,uart-mcumgr = &flexcomm3; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:30 */

22 };

23

24 /* node '/aliases' defined in deps/zephyr/dts/common/skeleton.dtsi:13 */

25 aliases {

26 led0 = &green_led; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:14 */

27 watchdog0 = &wwdt; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:15 */

28 usart-0 = &flexcomm3; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:16 */

29 i2c-0 = &flexcomm2; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:17 */

30 pwm-0 = &sctimer; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:18 */

31 sw0 = &user_button_0; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:19 */

32 ambient-temp0 = &p3t1755; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:20 */

33 };

34

35 /* node '/soc' defined in deps/zephyr/dts/arm/armv8-m.dtsi:6 */

36 soc {

PROBLEMS 6

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 3

GIT LENS

GIT LENS

zsh

-- west build: running target menuconfig

[0/1] cd /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/ze...ig.py /home/dms/projects/own/iomico/internal/demo/zephyr-cour

se/app/Kconfig

Loaded configuration '/home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/.config'

Configuration saved to '/home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/.config'

~/p/o/io/i/d/zephyr-course > #l2-demo ?4

~/p/o/io/i/d/zephyr-course > #l2-demo ?4

51s < .zephyr

51s < .zephyr

20:05:02

Ln 20, Col 108 (17 selected)

Tab Size: 4

UTF-8

LF

{ } DeviceTree

no context

zephyr.dts - zephyr-course - Visual Studio Code

39K 421K 49% 23.6G 99% 1 04/08 20:41



Devicetree: Aliases



What are aliases?



Named shortcuts to nodes — a stable public API for your app

```
/ {  
    aliases {  
        led0 = &blue_led;      // by node label  
        led1 = "/leds/led_red"; // by path  
    };  
};
```



Accessing nodes from C code



Three ways to get a node reference:

```
// By alias (preferred – stable API)
#define LED_NODE DT_ALIAS(led0)

// By node label (when no alias exists)
#define LED_NODE DT_NODELABEL(blue_led)

// By path (rarely used)
#define LED_NODE DT_PATH(leds, led_name)
```



Demo: Add Missing LEDs



The board DTS only defines a green LED. Add blue and red LEDs via overlay.

1. Create `boards/frdm_rw612.overlay` with blue and red LED nodes
2. Override `led0` alias to point to blue LED
3. Access the LED using `DT_ALIAS(led0)` – verify it blinks

Tip: Check `dts/bindings/gpio/gpio-leds.yaml` to see which properties are available

Tip: Check pin assignment in the DK schematic

FileEditSelectionViewGoRunTerminalHelp

ZEPHYR-COURSE

> .cache

> app

> boards

frdm_rw612.overlay

src

main.cpp

CMakeLists.txt

Kconfig

Kconfig.test

prj.conf

build

app

CMakeFiles

Kconfig

modules

zephyr

arch

boards

cmake

CMakeFiles

drivers

include

kconfig

kernel

lib

misc

soc

subsys

.cmake.dotconfig.checksum

.config

cmake_install.cmake

edt.pickle

edt.pickle.cmake

isr_tables_swi.ld

isr_tables_vt.ld

isr_tables.c

main.cpp 5, M

frdm_rw612.overlay U

zephyr.dts

Kconfig U

Kconfig.test U

app > boards > frdm_rw612.overlay

green_led

Aa ab .*

No results

1

2

3

4

5

6

7

8

9

10

11

12

/{

... leds {

... red_led: led_0 {

... gpio0 = <hsgpio0 1 GPIO_ACTIVE_LOW>;

... };

... blue_led: led_2 {

... gpio0 = <hsgpio0 0 GPIO_ACTIVE_LOW>;

... };

};

PROBLEMS 12

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 7

GITLENS

GITLENS

RAM: 9064 B 960 KB 0.92%

SMU1: 510 KB 510 KB 100.00%

SMU2: 140 KB 140 KB 100.00%

IDT_LIST: 0 GB 32 KB 0.00%

Generating files from /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.elf for board: frdm_rw612

~/p/o/io/i/d/zephyr-course > #l2-demo ?6

27s < .zephyr < 20:53:25

I2-demo*

Launchpad

4 1 7

Live Share

Update available, click to download.

Ln 4, Col 50

Spaces: 4

UTF-8

LF

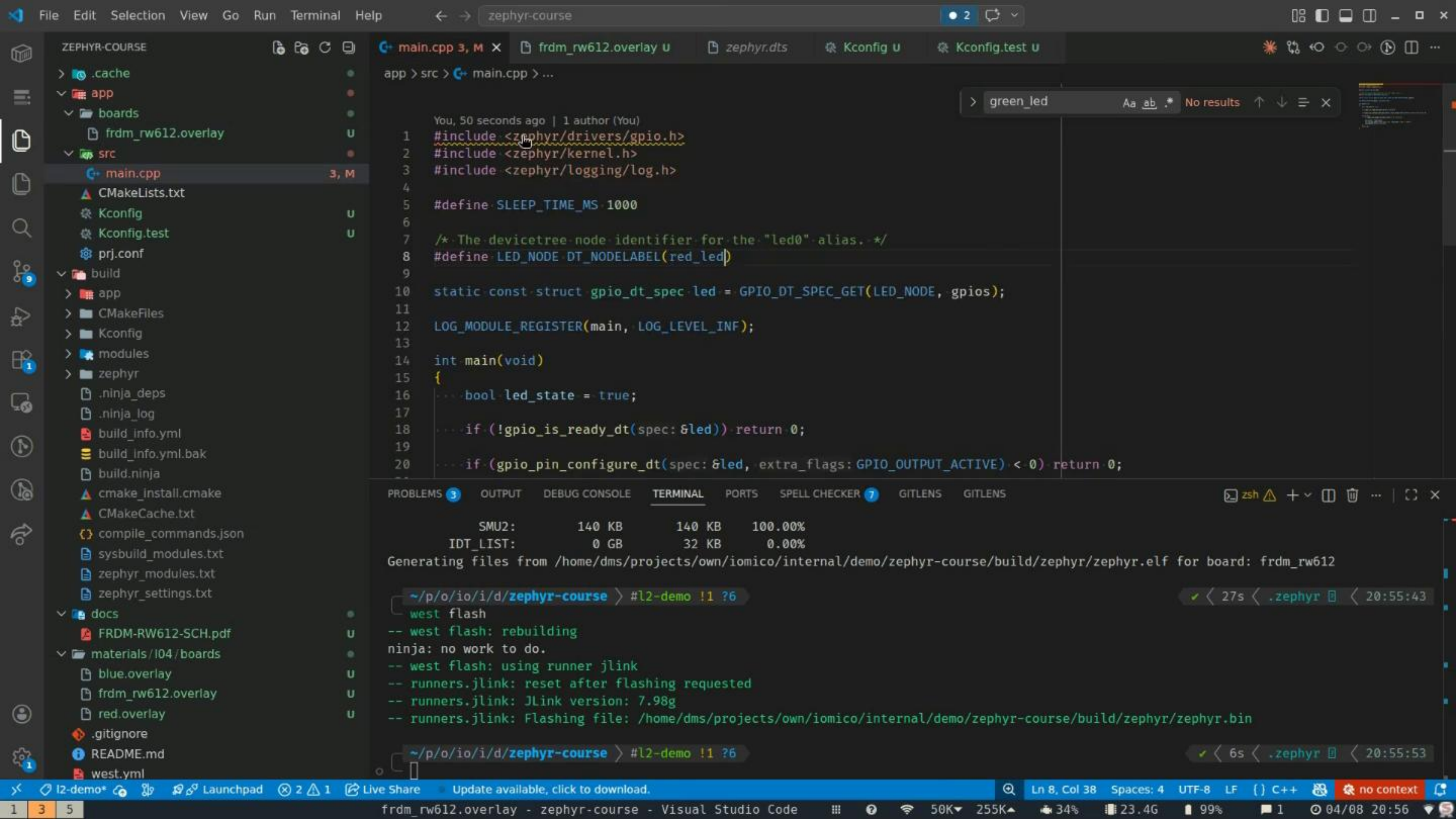
{ } Plain Text

no context

frdm_rw612.overlay - zephyr-course - Visual Studio Code

48K 259K 94% 22.7G 99%

04/08 20:55



The screenshot shows the Visual Studio Code interface for the Zephyr course project. The left sidebar displays the file explorer with the 'zephyr' directory expanded, showing files like 'edt.pickle.cmake', 'linker_zephyr_pre0.cmd', and 'zephyr.dts'. The main editor shows the 'zephyr.dts' file with a search for 'green_led' and a list of results. The bottom terminal shows the output of the 'west flash' command, indicating a successful flash operation.

```
build > zephyr > zephyr.dts
4 / {
1043
1044
1045 /* node '/leds' defined in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:33 */
1046 leds {
1047     compatible = "gpio-leds"; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:34 */
1048
1049     /* node '/leds/led_1' defined in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:35 */
1050     green_led: led_1 {
1051         gpios = < &hsgpio0 0xc 0x1 >; /* in deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612_common.dtsi:36 */
1052     };
1053
1054     /* node '/leds/led_0' defined in zephyr-course/app/boards/frdm_rw612.overlay:3 */
1055     red_led: led_0 {
1056         gpios = < &hsgpio0 0x1 0x1 >; /* in zephyr-course/app/boards/frdm_rw612.overlay:4 */
1057     };
1058
1059     /* node '/leds/led_2' defined in zephyr-course/app/boards/frdm_rw612.overlay:7 */
1060     blue_led: led_2 {
1061         gpios = < &hsgpio0 0x0 0x1 >; /* in zephyr-course/app/boards/frdm_rw612.overlay:8 */
1062     };
1063 };
1064
```

PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 7 GITLENS GITLENS

SMU2: 140 KB 140 KB 100.00%
IDT_LIST: 0 GB 32 KB 0.00%

Generating files from /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.elf for board: frdm_rw612

```
~/p/o/io/i/d/zephyr-course > #l2-demo !1 ?6
west flash
-- west flash: rebuilding
ninja: no work to do.
-- west flash: using runner jlink
-- runners.jlink: reset after flashing requested
-- runners.jlink: JLink version: 7.98g
-- runners.jlink: Flashing file: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.bin
```

~/p/o/io/i/d/zephyr-course > #l2-demo !1 ?6

FileEditSelectionViewGoRunTerminalHelp

ZEPHYR-COURSE

> .cache

> app

> boards

frdm_rw612.overlayU

src

main.cpp3, M

CMakeLists.txt

KconfigU

Kconfig.testU

prj.conf

build

app

CMakeFiles

Kconfig

modules

zephyr

arch

boards

cmake

CMakeFiles

drivers

include

kconfig

kernel

lib

misc

soc

subsys

.cmake.dotconfig.checksum

.config

cmake_install.cmake

edt.pickle

edt.pickle.cmake

isr_tables_swi.ld

isr_tables_vt.ld

isr_tables.c

main.cpp 3, M

frdm_rw612.overlay U

zephyr.dts

Kconfig U

Kconfig.test U

app > boards > frdm_rw612.overlay

> green_led

Aa ab .*

No results

↑ ↓ ≡ ×

```
1  /- {
2      leds {
3          red_led: led_0 {
4              gpios = <8hsgpio0 1 GPIO_ACTIVE_LOW>;
5          };
6
7          blue_led: led_2 {
8              gpios = <8hsgpio0 0 GPIO_ACTIVE_LOW>;
9          };
10     };
11 };
12
```

PROBLEMS 10

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 7

GITLENS

GITLENS

python3 ⚠ + ▾ 🗑 ... | 🔍 ×

west flash

-- west flash: rebuilding

[7/7] Linking CXX executable zephyr/zephyr.elf

Memory region

Used Size

Region Size

%age Used

FLASH: 39316 B 64 MB 0.06%

RAM: 9064 B 960 KB 0.92%

SMU1: 510 KB 510 KB 100.00%

SMU2: 140 KB 140 KB 100.00%

IDT_LIST: 0 GB 32 KB 0.00%

Generating files from /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.elf for board: frdm_rw612

-- west flash: using runner jlink

-- runners.jlink: reset after flashing requested

-- runners.jlink: JLink version: 7.98g

-- runners.jlink: Flashing file: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.bin

Ln 7, Col 24 (5 selected)

Spaces: 4

UTF-8

LF

{ } Plain Text

no context

1 3 5

main.cpp - zephyr-course - Visual Studio Code

48K 241K 56% 23.1G 99%

04/08 20:56

FileEditSelectionViewGoRunTerminalHelp

zephyr-course

2

ZEPHYR-COURSE

.cache

app

boards

frdm_rw612.overlay

src

main.cpp3, M

CMakeLists.txt

Kconfig

Kconfig.test

prj.conf

build

app

CMakeFiles

Kconfig

modules

zephyr

arch

boards

cmake

CMakeFiles

drivers

include

kconfig

kernel

lib

misc

soc

subsys

.cmake.dotconfig.checksum

.config

cmake_install.cmake

edt.pickle

edt.pickle.cmake

isr_tables_swi.ld

isr_tables_vt.ld

isr_tables.c

main.cpp3, M

frdm_rw612.overlay

zephyr.dts

Kconfig

Kconfig.test

app > src > main.cpp > ...

You, now | 1 author (You)

1#include <zephyr/drivers/gpio.h>

2#include <zephyr/kernel.h>

3#include <zephyr/logging/log.h>

4

5#define SLEEP_TIME_MS 1000

6

7/* The devicetree node identifier for the "led0" alias. */

8// /leds/led_2

9#define LED_NODE DT_PATH(leds, led_2)

10

11static const struct gpio_dt_spec led = GPIO_DT_SPEC_GET(LED_NODE, gpios);

12

13LOG_MODULE_REGISTER(main, LOG_LEVEL_INF);

14

15int main(void)

16{

17 bool led_state = true;

18

19 if (!gpio_is_ready_dt(&led)) return 0;

20

green_led

Aa ab

No results

PROBLEMS10

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER7

GITLENS

GITLENS

python3

west flash

-- west flash: rebuilding

[7/7] Linking CXX executable zephyr/zephyr.elf

Memory regionUsed SizeRegion Size%age Used

FLASH:39316 B64 MB0.06%

RAM:9064 B960 KB0.92%

SMU1:510 KB510 KB100.00%

SMU2:140 KB140 KB100.00%

IDT_LIST:0 GB32 KB0.00%

Generating files from /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.elf for board: frdm_rw612

-- west flash: using runner jlink

-- runners.jlink: reset after flashing requested

-- runners.jlink: JLink version: 7.98g

-- runners.jlink: Flashing file: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.bin

12-demo*

Launchpad

2

1

7

clangd: idle

Live Share

Update available, click to download.

Ln 9, Col 38

Spaces: 4

UTF-8

LF

{ } C++

no context

frdm_rw612.overlay - zephyr-course - Visual Studio Code

48K

241K

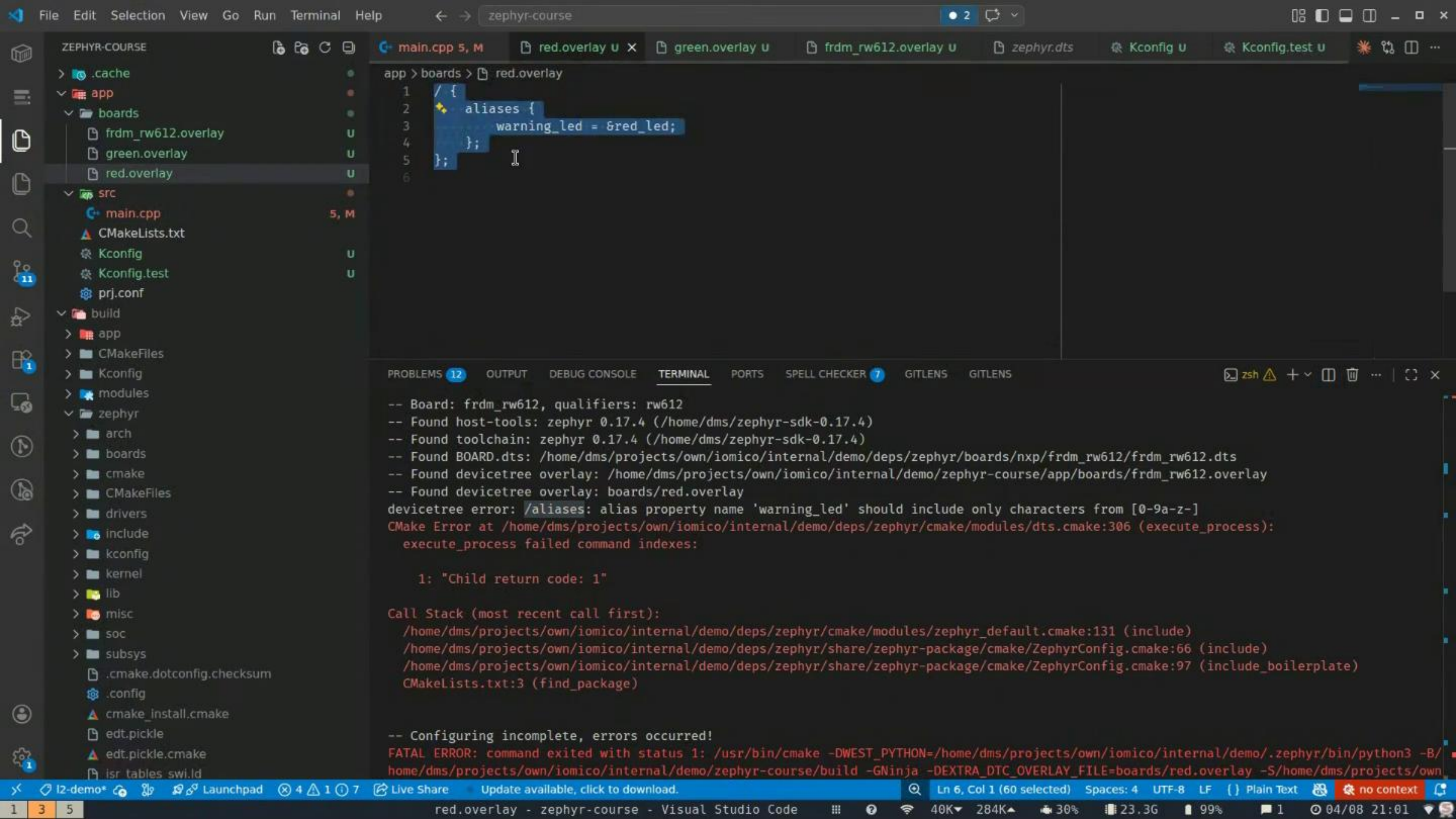
39%

23.1G

99%

1

04/08 20:56



FileEditSelectionViewGoRunTerminalHelp

ZEPHYR-COURSE

> .cache

> app

> boards

frdm_rw612.overlay

green.overlay

red.overlay

> src

main.cpp

CMakeLists.txt

Kconfig

Kconfig.test

prj.conf

> build

> app

> CMakeFiles

> Kconfig

> modules

> zephyr

> arch

> boards

> cmake

> CMakeFiles

> drivers

> include

> kconfig

> kernel

> lib

> misc

> soc

> subsys

.cmake.dotconfig.checksum

.config

cmake_install.cmake

edt.pickle

edt.pickle.cmake

isr_tables.swi.ld

main.cpp 3, M

red.overlay U

green.overlay U

frdm_rw612.overlay U

zephyr.dts

Kconfig U

Kc

app > src > main.cpp > ...

You, now | 1 author (You)

1#include <zephyr/drivers/gpio.h>

2#include <zephyr/kernel.h>

3#include <zephyr/logging/log.h>

4

5#define SLEEP_TIME_MS 1000

6

7/* The devicetree node identifier for the "led0" alias. */

8#define LED_NODE DT_ALIAS(warning_led)

9

10static const struct gpio_dt_spec led = GPIO_DT_SPEC_GET(LED_NODE, gpios);

11

12LOG_MODULE_REGISTER(main, LOG_LEVEL_INF);

13

14int main(void)

15{

PROBLEMS 10

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER 7

GIT LENS

GIT LENS

python3

-- Configuring done (7.7s)

-- Generating done (0.1s)

-- Build files have been written to: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build

-- west build: building application

[152/152] Linking CXX executable zephyr/zephyr.elf

Memory regionUsed SizeRegion Size%age Used

FLASH:39316 B64 MB0.06%

RAM:9064 B960 KB0.92%

SMU1:510 KB510 KB100.00%

SMU2:140 KB140 KB100.00%

IDT_LIST:0 GB32 KB0.00%

Generating files from /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.elf for board: frdm_rw612

~/p/o/io/i/d/zephyr-course > #l2-demo !1 ?6

west flash

-- west flash: rebuilding

ninja: no work to do.

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-- runners.jlink: Flashing file: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.bin

12-demo*

Launchpad

2

1

7

clangd: idle

Live Share

Update available, click to download.

You, now

Not Committed Yet

Ln 8, Col 39

Spaces: 4

UTF-8

LF

{ } C++

no context

main.cpp - zephyr-course - Visual Studio Code

43K

325K

49%

23.0G

99%

1

04/08 21:04



Homework: Kconfig + Overlay Integration



Homework: Heartbeat LED via Kconfig + Overlay



Add a configurable heartbeat LED to the blinky app from the demo. Works on any board.

- Create `app.overlay` — add alias `app-led` pointing to your board's `led0`
- Add `Kconfig` file with;
 - `int APP_HEARTBEAT_PERIOD_MS` (default `500`, range `100` — `2000`)
- In C: use `DT_ALIAS(app_led)` for the GPIO and `CONFIG_APP_HEARTBEAT_PERIOD_MS` for the sleep duration
- Verify: open `menuconfig`, change the period, rebuild — LED blink speed must change
- Push tag: `l4-task1`



Best Practices



Devicetree



- ✓ Use **aliases** for your app's public API (`led0` , `warning_led`)
- ✓ Keep overlays in `boards/` – committed to git
- ✓ Label nodes you'll reference from code
- ✓ Use `compatible` to match the right driver
- ✗ Don't hardcode GPIO pins in C code
- ✗ Don't commit generated DTS (`build/zephyr/zephyr.dts`)

Thank You!

Questions?

